

Online Appendix

Coronavirus Perceptions and Economic Anxiety

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This Online Appendix contains supplementary materials mentioned in the main text. Section A provides an overview of the methodology underlying the analysis of the impact of state-level coronavirus arrival on outcomes in the US. In Appendix Section B we display additional figures. Online Appendix Figure A.1 displays the global trends in search intensity for our four main indicators. Online Appendix Figure A.2 displays the impact of dropping individual countries from the main search intensity analysis. Online Appendix Figure A.3 displays the impact of dropping subregions from the main search intensity analysis. Online Appendix Figure A.4 displays how perceptions of severity and economic worries change over time for different subgroups. Online Appendix Figure A.5 displays search intensity trends in the US and globally for our four main measures. Online Appendix Figure A.6 displays changes in economic expectations from early to mid-March using Roper Center polling data. Online Appendix Figure A.7 plots the non-parametric relationship between mortality and contagiousness perceptions and economic anxieties. Finally, Online Appendix Figure A.8 displays the results of categorizing individuals in linear, exponential, and other mental models of disease spread. It also shows the correlation between mental models and economic anxieties.

Online Appendix Section C displays additional tables. Online Appendix Table A.1 contains all countries used for the Google search intensity analysis. Online Appendix Table A.2 displays correlations of Google search intensity with GDP and its components. Online Appendix Table A.3 contains the main Google search intensity results also displayed in Figure 1. Online Appendix Table A.4 displays the main results without normalization of the outcome variables. Online Appendix Table A.5 displays a series of placebo difference-in-differences regressions. Online Appendix Table A.6 displays summary statistics for the March 5 survey. Online Appendix Table A.7 displays summary

statistics for the March 16 survey. Online Appendix Table A.8 shows the correlation between having any coronavirus case and coronavirus-related concerns in mid February. Online Appendix Table A.9 shows differences in coronavirus perceptions and economic anxieties over time. Online Appendix Table A.10 show the results of the difference-in-differences regression described in Section A. Online Appendix Tables A.11 and A.12 display correlations of coronavirus perceptions and economic anxieties for binary and continuous variables, respectively. Online Appendix Table A.13 shows balance tests for both experiments. Online Appendix Table A.14 shows the impact of information about relative mortality on the perceived severity of the crisis. Online Appendix Table A.15 shows the impact of coronavirus-related information on economic worries. Online Appendix Table A.16 shows the impact of contagiousness information controlling for treatment assignment in the relative mortality experiment. Online Appendix Table A.17 explores interactions effects between the experiments. Online Appendix Table A.18 displays predictors of mental models of disease spread. Finally, Online Appendix Table A.19 displays the correlations between predicted disease spread and severity perceptions as well as economic worries.

A The Impact of Coronavirus Arrival in the US

In this section, we describe the methodology for the additional analysis linking confirmed coronavirus cases in the US to the perceived threat of the pandemic and economic anxieties. For this purpose, we leverage a public opinion poll and our own survey data.

A.1 Cross-sectional Evidence from the US in mid-February

The US reported its first case of coronavirus on January 22, 2020 (Dong et al., 2020). For most of February, the case count within the US remained fairly flat, increasing from 8 cases on February 1st to 24 cases by February 29. We present descriptive evidence documenting that there is an association between the spread of coronavirus within the US and increased anxieties using opinion polling data from individuals across US states. The Kaiser Family Foundation poll was conducted from February 13 to February 18, 2020 among a sample of 1207 US residents and included a few questions relating to coronavirus.¹⁷

During that whole period, there were no reported new cases of coronavirus across the US with the total confirmed case count staying flat at 13 cases. 46 states reported no case. Three states (Washington, Massachusetts and Arizona) reported a single case each, Illinois reported two cases and California reported eight cases. We assess associations between residing in a state with at least any coronavirus case and responses to coronavirus-related questions. To do so, we estimate the following simple regression:

$$y_{i,s,t} = \gamma \times anycase_s + \beta' X_i + \eta_t + \epsilon_{is}$$

where $anycase_s$ is a dummy indicating the presence of at least one coronavirus case.

As indicated, there was no further spread recorded during the sample period according to data from Dong et al. (2020).¹⁸

¹⁷The underlying micro data are made available through the Roper Center ID 31117209.

¹⁸The fact that the US reported little intracommunity spread during much of February is likely not the result of no spread occurring, but rather due to the failure of the US to ramp up testing, and the use of a faulty test, see [nature.com/articles/d41586-020-01068-3](https://www.nature.com/articles/d41586-020-01068-3).

The dependent variable $y_{i,s,t}$ measures a survey respondent's responses to a set of coronavirus related questions: whether individuals are (very) concerned or (not at all) concerned about: the family or oneself getting sick, a negative economic impact, or a widespread outbreak of coronavirus in the US.

We study whether individuals living in states with any coronavirus cases during the time period give different responses. We control for interview date fixed effects, η_t , along with a set of individual-level controls.

Results The results are presented in Table A.8 and suggest that respondents from states with any case of coronavirus during that sample period are more concerned about: themselves or family members getting sick; the negative impact on the US economy; and about a widespread outbreak of coronavirus in the US.

A.2 Difference-in-differences Analysis in March

To go beyond a purely cross-sectional analysis of the relationship between the presence of coronavirus cases and economic anxieties we conduct a state-level difference-in-differences analysis. We use the fact that in the 11-day period between our surveys on March 5 and 16, 31 states recorded their first coronavirus case leaving only three states without a case on March 15. This allows us to estimate a regression of the following form:

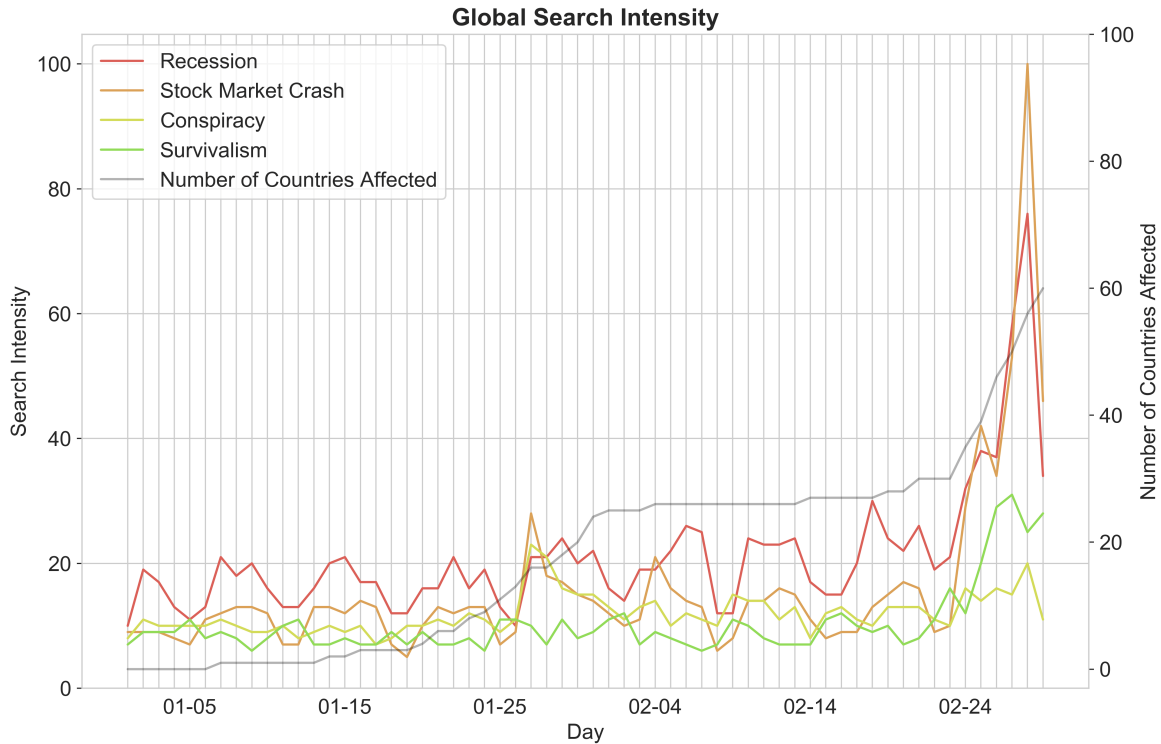
$$y_{ist} = \alpha \times anycase_{st} + \delta_s + \eta_t + \beta' X_i + \varepsilon_{ist} \quad (4)$$

where $anycase_{st}$ is a dummy variable indicating whether a given state has a confirmed coronavirus case by the time of the survey. We also include state fixed effects to account for permanent differences in economic anxieties across states. Finally, time fixed effects account for any level differences across time that affect all states in the same way. We conduct this analysis for the main four outcomes measuring perceptions of the severity of the pandemic and worries about its impact on the economy.

Results Online Appendix Table A.10 displays estimation results. We find that individuals in states with at least one coronavirus case exhibit a significantly higher level of perceived threat by the pandemic and worries about the US economy. The relationship impact on worries about personal economic circumstances is also positive but it is smaller and not statistically significant.

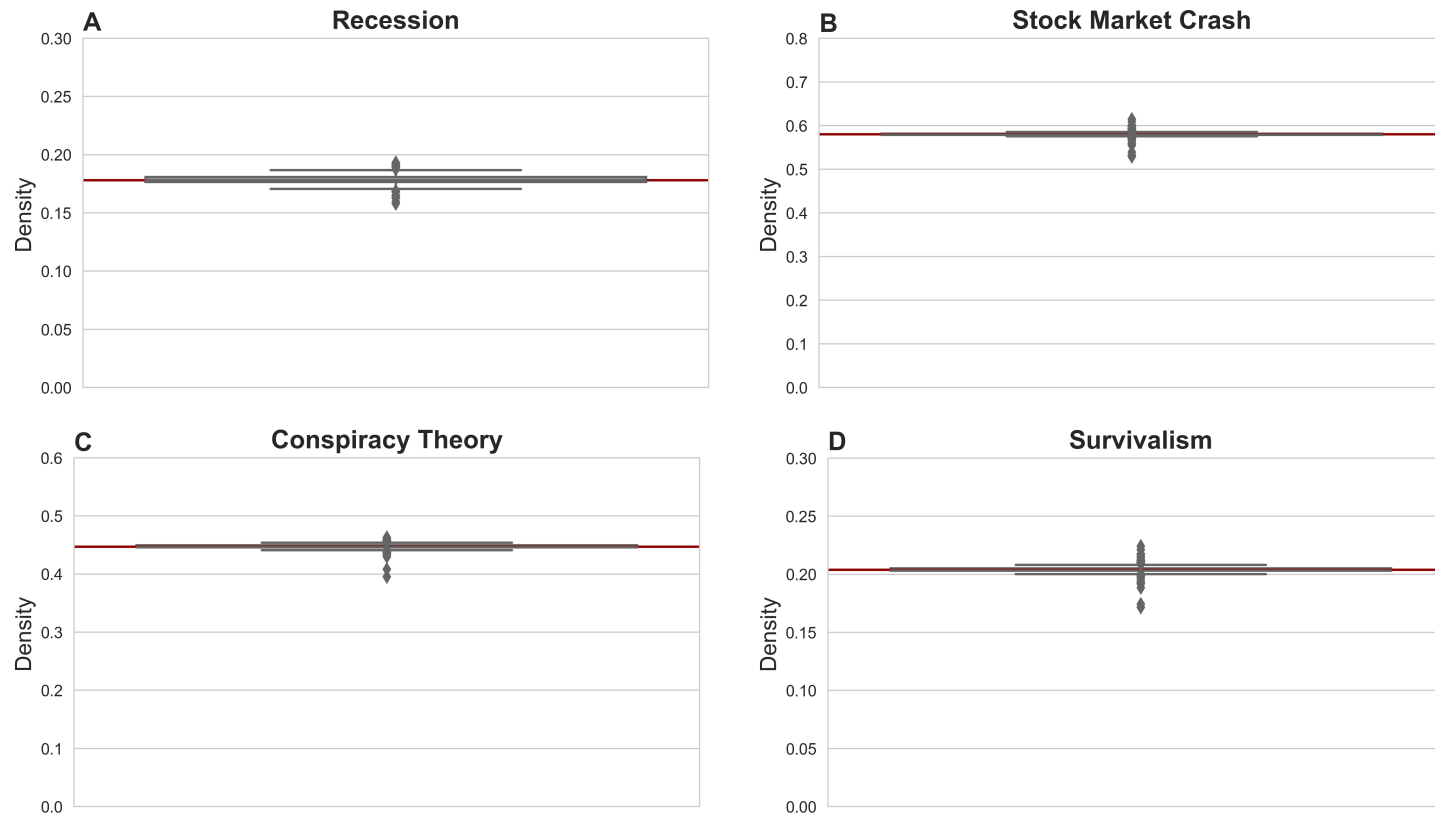
B Online Appendix Figures

Figure A.1: Time Series of Global Internet Searches for January and February 2020



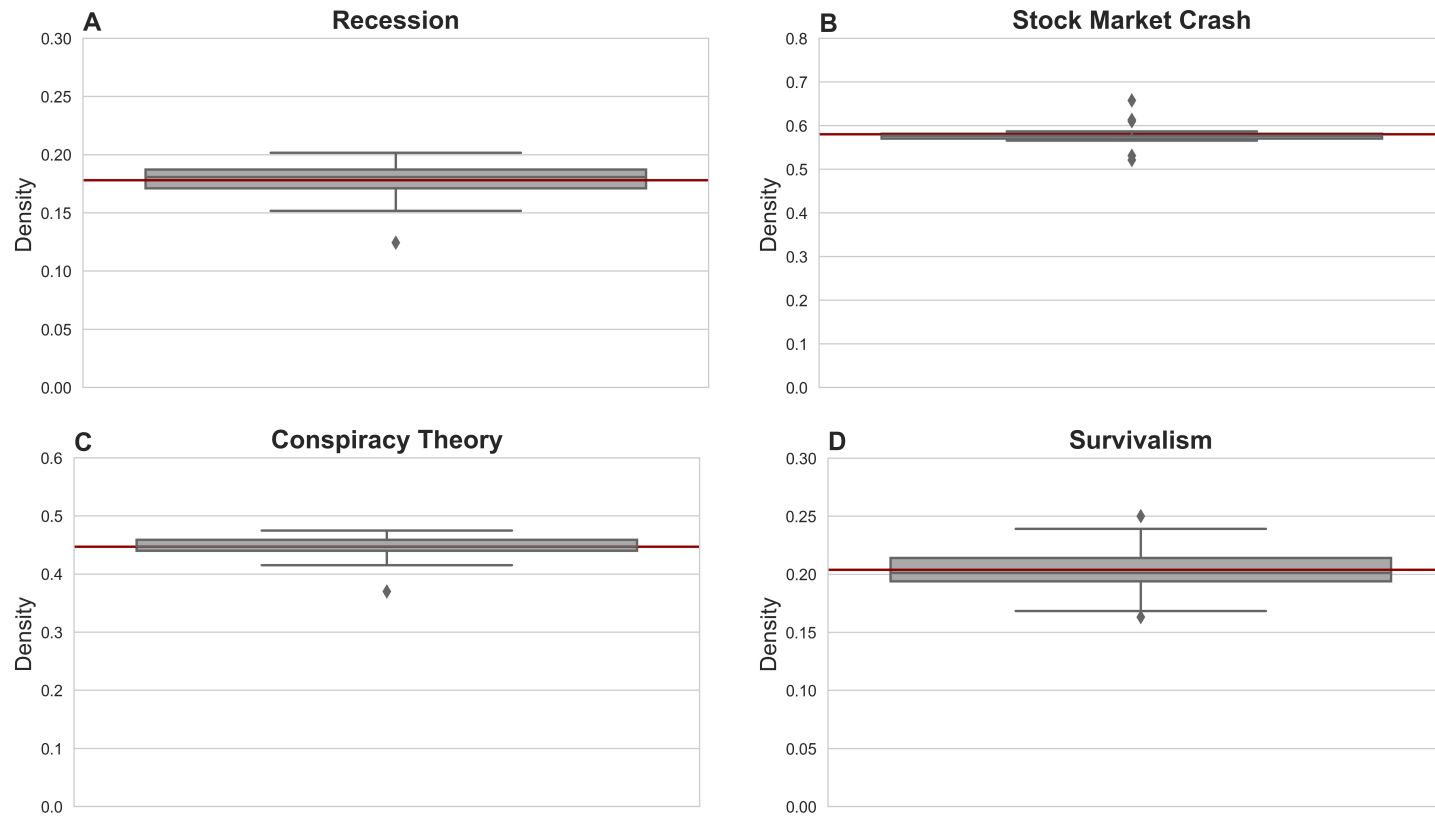
Notes: Online Appendix Figure A.1 shows the time series of the search intensity for the Google topics "Recession", "Stock Market Crash", "Conspiracy Theory", and "Survivalism" from January 1st to February 29th, 2020 as well as the number of countries with a confirmed coronavirus case.

Figure A.2: Robustness of results to dropping individual countries



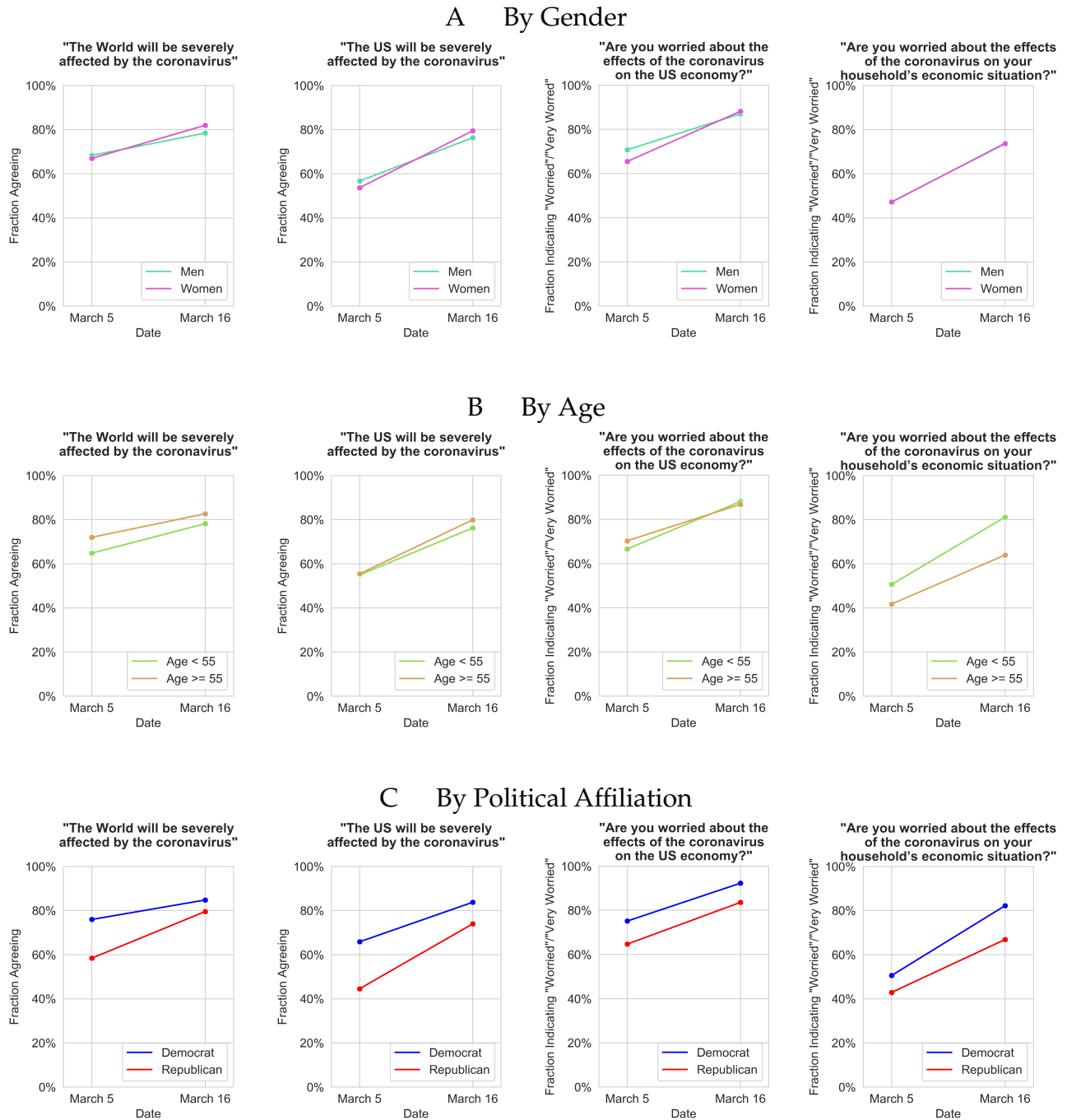
Notes: Online Appendix Figure A.2 shows boxplots of the point estimates obtained from estimating equation 1 when dropping each country in turn. The coefficient obtained from estimation on the full sample is indicated by the horizontal red line.

Figure A.3: Robustness of results to dropping individual sub-regions



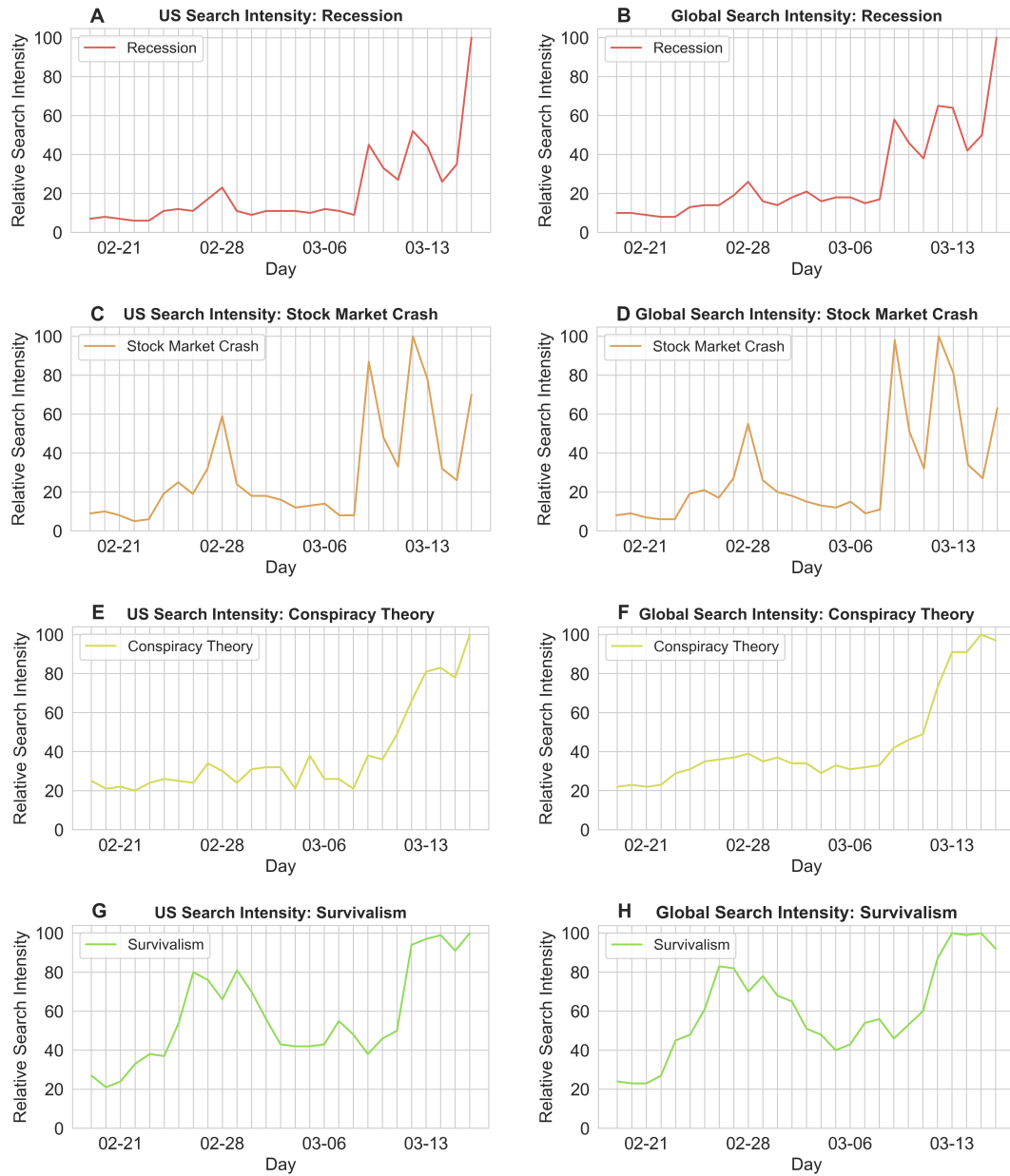
Notes: Online Appendix Figure A.3 shows boxplots of the point estimates obtained from estimating equation 1 when dropping countries belonging to each of the 17 different sub-regions in turn. The coefficient obtained from estimation on the full sample is indicated by the horizontal red line.

Figure A.4: Evolution of Beliefs about Severity of Crisis and Economic Worries by Subgroups



Notes: Online Appendix Figure A.4 shows for different subgroups the evolution of beliefs about the severity of the crisis for the world (leftmost panels) and the United States (second leftmost panels) as well as worries about the aggregate economy (second rightmost panels) and worries about respondents' personal economic situation (rightmost panels). The data were collected on March 5 and March 16, 2020.

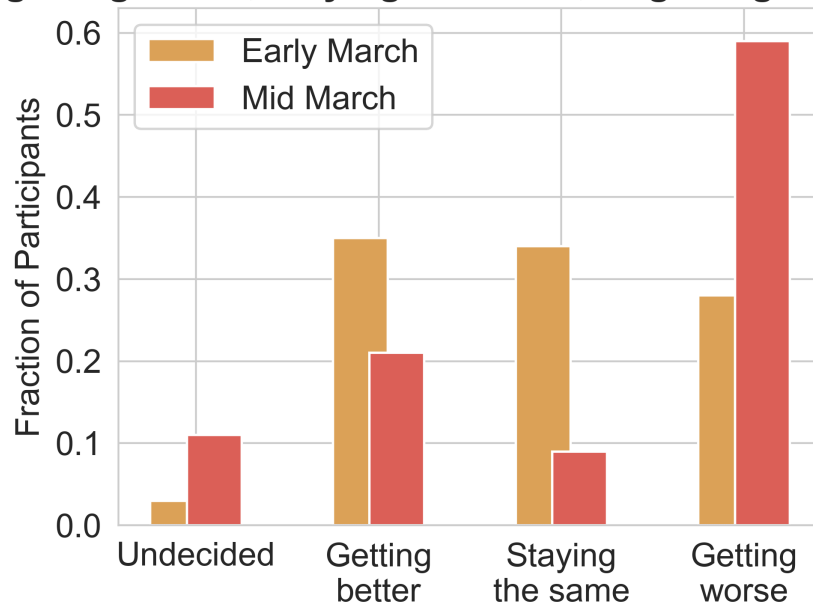
Figure A.5: Time Series Google Search Intensity for the United States and Worldwide from 02-19-2020 to 03-16-2020.



Notes: Online Appendix Figure A.5 shows time series of the search intensity for Google topics "Recession", "Stock Market Crash", "Conspiracy Theory", and "Survivalism" from February 19th to March 16th, 2020 for the United States and worldwide.

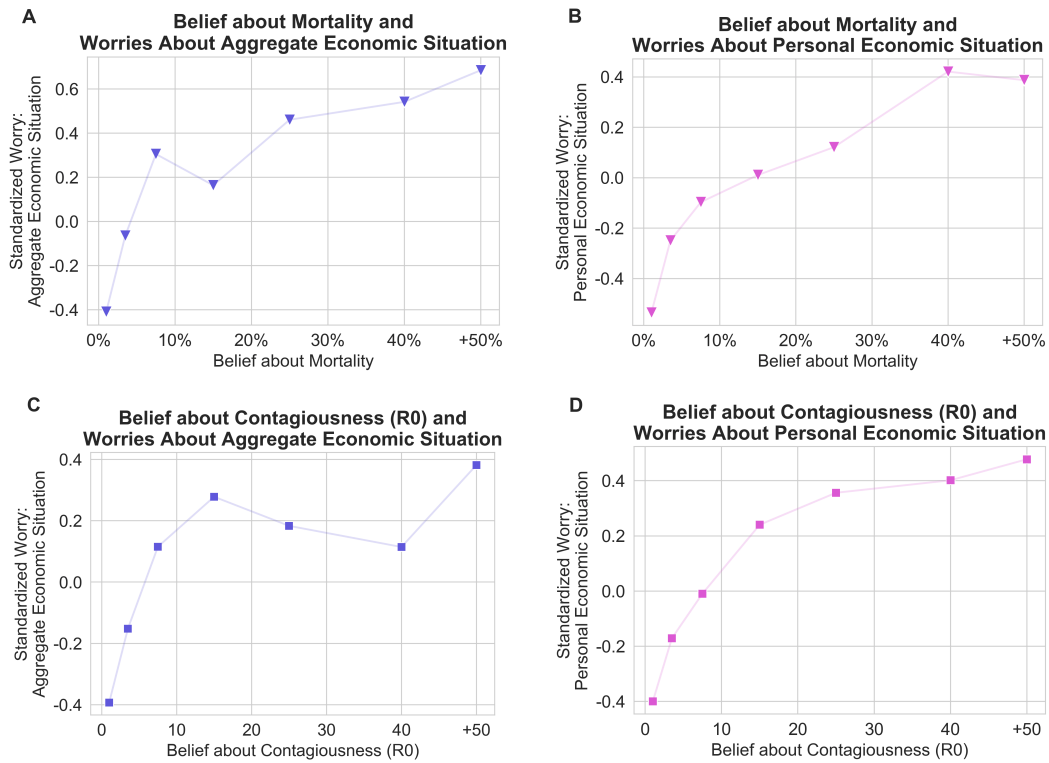
Figure A.6: Change in economic outlook in opinion polls

Survey: "Do you think the national economy is getting better, staying the same, or getting worse?"



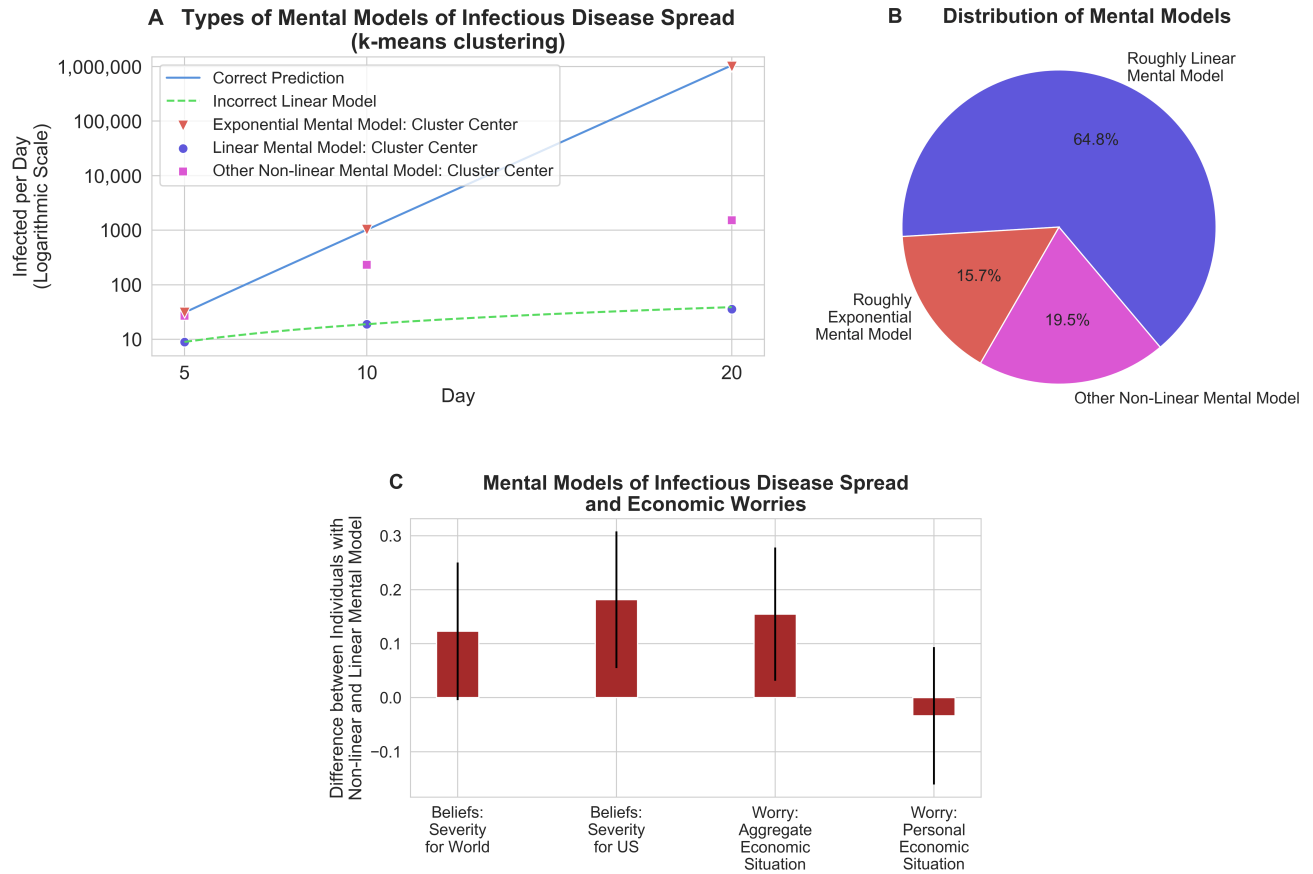
Notes: Online Appendix Figure A.6 compares polling results among representative samples of the US population from an opinion poll conducted between March 5 and 8 2020 (American Research Group Poll, Question 4, Roper Center Poll Identifier: 31117199.00003) with results from an opinion poll conducted between March 16 and 19 2020 (Quinnipiac University Poll, Question 19, Roper Center Poll Identifier: 31 31117223.00023). The Figure displays answers to the following question in both opinion polls: "Do you think the national economy is getting better, staying the same, or getting worse?"

Figure A.7: Beliefs about Mortality and Contagiousness (R0) and Economic Anxieties



Notes: Online Appendix Figure A.7 displays the non-parametric relationship between perceptions of the novel coronavirus and economic anxieties. The data were collected on March 5. Panel A and B show non-parametric relationships between respondents' worries about the aggregate and personal economic situation and their belief about coronavirus mortality. Panel C and D show non-parametric relationships between respondents' worries about the aggregate and their personal economic situation and their beliefs about coronavirus contagiousness (R0). Each plot shows a binscatter plot where x-values correspond to the midpoint of each bin. y-values indicate the mean of the outcome variable for the respective bin.

Figure A.8: Classification of Mental Models of Infectious Disease Spread



Notes: Online Appendix Figure A.8 shows a classification of mental models of infectious disease spread obtained from k-means clustering using 3 clusters in the space of predicted log-number of cases at day 5, 10, and 20. Panel A shows the median number of predicted cases in the three clusters (dots) as well as the cluster centers (shaded areas) that contain 50% of the mass of each cluster. The blue and green lines represent the correct prediction values as well as an incorrect linear model with a slope of 2. The three types obtained from the cluster approach can be described as a correct 'exponential mental model', an incorrect 'linear mental model', or an incorrect 'other mental model'. Panel B visualizes the type distribution in the data. Panel C shows estimates of OLS regressions with an indicator for exhibiting a non-linear mental model as the independent variable and using as the dependent variables participants' beliefs about the crisis' severity for the world and the United States as well as their worries about the aggregate economy and their personal economic situation. Error bars indicate 95% confidence intervals.

C Online Appendix Tables

Table A.1: Countries and territories included in the analysis of Google searches

Andorra	Dominica	Korea, Republic of ^{abc}	Palestine, State of
United Arab Emirates ^b	Dominican Republic	Kuwait ^b	Portugal ^{abc}
Afghanistan	Algeria ^b	Cayman Islands	Paraguay
Antigua and Barbuda	Ecuador ^{ac}	Kazakhstan ^{ab}	Qatar ^{ab}
Albania	Estonia ^{ab}	Laos	Reunion
Armenia	Egypt ^{ac}	Lebanon	Romania ^{abc}
Angola	Spain ^{abc}	Saint Lucia	Serbia ^{abc}
Argentina ^{abc}	Ethiopia	Sri Lanka ^b	Russian Federation ^{abc}
Austria ^{abc}	Finland ^{abc}	Liberia	Rwanda
Australia ^{abc}	Fiji	Lesotho	Saudi Arabia ^a
Aruba	Faroe Islands	Lithuania ^{abc}	Sudan
Azerbaijan ^{ab}	France ^{abc}	Luxembourg	Sweden ^{abc}
Bosnia and Herzegovina	Gabon	Latvia ^{abc}	Singapore ^{abc}
Barbados	United Kingdom ^{abc}	Libya	Saint Helena
Bangladesh ^b	Grenada	Morocco ^b	Slovenia ^{abc}
Belgium ^{abc}	Georgia	Moldova, Republic of	Slovakia ^{abc}
Burkina Faso	French Guiana	Montenegro	Sierra Leone
Bulgaria ^{abc}	Guernsey	Madagascar	Senegal
Bahrain ^b	Ghana	North Macedonia	Somalia
Burundi	Gibraltar	Mali	Suriname
Benin	Greenland	Myanmar	El Salvador ^{ab}
Bermuda	Guinea	Mongolia	Sint Maarten (Dutch part)
Brunei Darussalam	Guadeloupe	Macao	Syrian Arab Republic
Bolivia (Plurinational State of)	Greece ^{abc}	Martinique	Eswatini
Bonaire, Sint Eustatius and Saba	Guatemala	Mauritania	Togo
Brazil ^{abc}	Guam	Malta	Thailand ^{abc}
Bahamas	Guyana	Mauritius	Tajikistan
Bhutan	Hong Kong ^{abc}	Maldives	Turkmenistan
Botswana	Honduras	Malawi	Tunisia ^{ab}
Belarus	Croatia ^{ab}	Mexico ^{abc}	Tonga
Belize	Haiti	Malaysia ^{abc}	Turkey ^{abc}
Canada ^{abc}	Hungary ^{abc}	Mozambique	Trinidad and Tobago
Congo, Democratic Republic of the	Indonesia ^{abc}	New Caledonia	Taiwan, Province of China ^{abc}
Congo	Ireland ^{abc}	Niger	Tanzania, United Republic of
Switzerland ^{abc}	Israel ^{abc}	Nigeria ^{abc}	Ukraine ^{abc}
Côte d'Ivoire	Isle of Man	Nicaragua	Uganda
Chile ^{abc}	India ^{abc}	Netherlands ^{abc}	United States of America ^{abc}
Cameroon	Iraq	Norway ^{abc}	Uruguay
China ^{ab}	Iran (Islamic Republic of) ^{abc}	Nepal	Uzbekistan
Colombia ^{abc}	Iceland	New Zealand ^{abc}	Saint Vincent and the Grenadines
Costa Rica ^{ac}	Italy ^{abc}	Oman	Venezuela (Bolivarian Republic of) ^{ac}
Cuba	Jersey	Panama	Virgin Islands (U.S.)
Cabo Verde	Jamaica	Peru ^{abc}	Viet Nam ^{ab}
Curacao	Jordan ^{ab}	French Polynesia	Yemen
Cyprus ^{abc}	Japan ^{abc}	Papua New Guinea	South Africa ^{abc}
Czechia ^{abc}	Kenya ^a	Philippines ^{abc}	Zambia
Germany ^{abc}	Kyrgyzstan	Pakistan ^b	Zimbabwe
Djibouti	Cambodia	Poland ^{abc}	
Denmark ^{abc}	Saint Kitts and Nevis	Puerto Rico	

Notes: Appendix Table A.1 lists all countries included in the analysis of the impact of the coronavirus on internet searches on Google. The superscripts *a, b, c* refer to data availability regarding aggregate demand components for the analysis in Online Appendix Table A.2: *a* data available on real GDP; *b* data available on industrial production; *c* data available on demand factors.

Table A.2: Increases in ‘Recession’ topic Google searches are a leading indicator of subsequent aggregate demand contractions

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Real GDP	Industrial production	C	Demand factors			
				I	G	X	M
L.Recession topic Google searches	-1.009*** (0.311)	-1.231* (0.661)	-1.564*** (0.506)	-1.847 (1.798)	-1.345 (0.888)	1.109* (0.657)	-5.063*** (1.352)
R ²	.716	.446	.627	.27	.282	.236	.314
Countries	70	72	58	58	58	58	58
Observations	1350	1218	1087	1087	1087	1087	1087
Country FE	X	X	X	X	X	X	X
Year x Quarter FE	X	X	X	X	X	X	X

Notes: Appendix Table A.2 displays the relationship between year-on-year growth rates in GDP, industrial production and demand components and ‘Recession’ topic Google searches. Econometrically, we perform country-level regressions controlling for country and year-by-quarter fixed effects in all specifications. The results show that increases in Google search activity for recession-related topics are associated with lower growth rates in GDP, consumption spending and imports in the subsequent quarter. The level of analysis is country and quarter. Data were collected by the Economist Intelligence Unit from 2015 to 2019. The dependent variable in column (1) measures GDP growth. The dependent variable in column (2) measures growth of industrial production. Columns (3) to (6) measure different components of aggregate demand. Column (3) shows the association with aggregate consumption. Column (4) shows the association with investments. Column (5) shows the association with government spending. Column (6) shows the association with exports. Column (7) shows the association with imports. The independent variable measures Google search intensity for the topic “recession”. For countries included in each regression, see Online Appendix Table A.1. Standard errors clustered at the country level are presented in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.3: The impact of coronavirus arrival on Google searches related to economic anxiety

	Impact on Goolge search trends			
	(1) Recession	(2) Stock Market Crash	(3) Conspiracy Theory	(4) Survivalism
Panel A: Any Covid-19 case				
Post any Covid-19 case	0.178** (0.073)	0.580*** (0.124)	0.447*** (0.091)	0.204*** (0.073)
R ²	0.04	0.10	0.06	0.13
Number of Observations	11640	11640	11640	11640
Panel B: Any human-to-human transmission				
Post any human-to-human transmission	0.351** (0.141)	0.293* (0.163)	0.388** (0.164)	0.354** (0.140)
R ²	0.04	0.10	0.06	0.13
Number of Observations	11640	11640	11640	11640
Number of countries	194	194	194	194
Country FE	X	X	X	X

Notes: Online Appendix Table A.3 displays the impact of coronavirus arrival on Google searches for search terms related to economic anxiety. The results show that coronavirus arrival is a predictor of Google searches related to economic anxiety. Column 1 shows results for Google searches related to recessions. Column 2 shows results for Google searches related to stock market crashes. Column 3 shows results for Google searches related to conspiracy topics. Column 4 shows results for Google searches related to survivalism. The dependent variable measures Google search intensity for the indicated topics normalized by the average search intensity in a country prior to the coronavirus arrival. The data on Google searches were downloaded from the Google API on March 3rd. In panel A, we show the impact of a dummy variable indicating at least one coronavirus case. In Panel B, we show the impact of having at least one human-to-human transmission of coronavirus. The data on first cases stem from Dong et al. (2020). The data on human-to-human transmissions are based on official reports by the WHO and national authorities. The level of analysis is country-day. Dates included range from January 1st 2020 to February 29th 2020. The table displays coefficients that are estimated using a linear regression model with country fixed effects and day fixed effects. Standard errors clustered at the country level are presented in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.4: Robustness of results to not normalizing – Impact of coronavirus arrival on Google searches related to economic anxiety

	Impact on Goolge search trends			
	(1) Recession	(2) Stock Market Crash	(3) Conspiracy Theory	(4) Survivalism
Panel A: Any Covid-19 case				
Post any Covid-19 case	2.522*** (0.677)	2.824*** (0.538)	3.252*** (0.580)	1.493*** (0.443)
R ²	0.27	0.26	0.31	0.36
Number of Observations	11640	11640	11640	11640
Panel B: Any human-to-human transmission				
Post any human-to-human transmission	5.644*** (1.927)	1.792* (1.069)	3.622* (1.999)	2.281** (0.978)
R ²	0.27	0.26	0.31	0.36
Number of Observations	11640	11640	11640	11640
Number of countries	194	194	194	194
Country FE	X	X	X	X

Notes: The dependent variable measure country-specific search intensity for a topic indicated in the column head from January 2020 to February 29 2020. Column 1 shows results for Google searches related to recessions. Column 2 shows results for Google searches related to stock market crashes. Column 3 shows results for Google searches related to conspiracy topics. Column 4 shows results for Google searches related to survivalism. The dependent variable measures Google search intensity for the indicated topics. The data on Google searches were downloaded from the Google API on March 3rd. In panel A, we show the impact of a dummy variable indicating at least one coronavirus case. In Panel B, we show the impact of having at least one human-to-human transmission of coronavirus. The data on first cases stem from Dong et al. (2020). The data on human-to-human transmissions are based on official reports by the WHO and national authorities. The level of analysis is country-day. Dates included range from January 1st 2020 to February 29th 2020. The table displays coefficients that are estimated using a linear regression model with country fixed effects and day fixed effects. Standard errors are clustered at the country are presented in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.5: The impact of coronavirus arrival on placebo Google searches

	Impact on google searches for									
	(1) Dog	(2) Horse	(3) Insect	(4) DaVinci	(5) Nelson Mandela	(6) Rain	(7) Rainbow	(8) Stars	(9) Mars (planet)	(10) Menstrual Cycle
Panel A: First confirmed case										
First confirmed case	-0.018 (0.018)	-0.036 (0.027)	-0.007 (0.042)	-0.044 (0.038)	-0.033 (0.058)	0.044 (0.048)	-0.042 (0.045)	-0.012 (0.033)	-0.001 (0.103)	0.033 (0.033)
R ²	0.05	0.03	0.02	0.04	0.03	0.04	0.03	0.02	0.06	0.04
Number of Observations	11640	11640	11639	11249	11483	11640	11640	11640	11507	11477
Panel B: First human-to-human transmission										
First human-to-human transmission	-0.021 (0.038)	-0.085 (0.064)	0.035 (0.040)	-0.030 (0.039)	0.027 (0.059)	-0.041 (0.078)	-0.047 (0.055)	-0.024 (0.048)	0.102 (0.324)	0.039 (0.028)
R ²	0.05	0.03	0.02	0.04	0.03	0.04	0.03	0.02	0.06	0.04
Number of Observations	11640	11640	11639	11249	11483	11640	11640	11640	11507	11477
Number of countries	194	194	194	193	194	194	194	194	194	193
Country FE	X	X	X	X	X	X	X	X	X	X
Day FE	X	X	X	X	X	X	X	X	X	X

Notes: Online Appendix Table A.5 displays the impact of coronavirus arrival on placebo Google searches that should not be affected by the arrival of the coronavirus. The results document that coronavirus arrival does not systematically predict Google searches unrelated to economic anxiety. The dependent variable measures Google search intensity for the indicated topics normalized by the average search intensity in a country prior to the coronavirus arrival. The data on Google searches were downloaded from the Google API on March 3rd. In panel A, we show the impact of a dummy variable indicating at least one coronavirus case. In Panel B, we show the impact of having at least one human-to-human transmission of coronavirus. The data stem from Dong et al. (2020). The data on human-to-human transmissions are based on official reports by the WHO and national authorities. The level of analysis is country-day. Dates included range from January 1st 2020 to February 29th 2020. The table displays coefficients that are estimated using a linear regression model with country fixed effects and day fixed effects. Standard errors clustered at the country level are presented in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.6: Summary statistics: Experimental sample March 5

	(1) Mean	(2) SD	(3) Median	(4) Obs.
Panel A: Demographics				
% Male	49.02	50.02		914
% Age < 35	0.23	0.42		914
% Highschool education	17.61	38.12		914
% College education	80.53	39.62		914
% Currently working	55.03	49.77		914
% Democrat	40.04	49.03		914
% Republican	33.15	47.10		914
% High trust in science	1.09	10.41		914
Panel B: Economic Anxieties				
% agree: world severely affected by coronavirus	67.61	46.82		914
% agree: US severely affected by coronavirus	55.14	49.76		914
% worried about US economy	68.05	46.65		914
% worried about personal econ. situation	47.16	49.95		914
Panel C: Coronavirus perceptions				
Infectiousness (R0)	43.23	146.17	10	914
Predicted mortality rate	13.70	20.84	5	914

Notes: Online Appendix Table A.6 displays summary statistics for the experimental sample. These data were collected on March 5. Panel A shows shares of respondents with indicated characteristics. Panel B shows shares of respondents with indicated beliefs about the severity of the crisis and economic anxieties. Panel C shows variables measuring perceptions of the coronavirus.

Table A.7: Summary statistics: Exponential growth survey March 16

	(1) Mean	(2) SD	(3) Median	(4) Obs.
Panel A: Demographics				
% Male	52.09	49.98		1006
% Age < 35	22.66	41.89		1006
% Highschool education	19.98	40.00		1006
% College education	76.64	42.33		1006
% Currently working	52.19	49.98		1006
% Democrat	38.57	48.70		1006
% Republican	32.11	46.71		1006
% High trust in science	1.79	13.26		1006
Panel B: Economic Anxieties				
% agree: world severely affected by coronavirus	80.12	39.93		1006
% agree: US severely affected by coronavirus	77.83	41.56		1006
% worried about US economy	87.57	33.00		1006
% worried about personal econ. situation	73.76	44.02		1006
Panel C: Coronavirus perceptions				
Infectiousness (R0)	49.81	175.13	5	1006
Number of cases after 5 days (w)	20.02	20.72	11	1006
Number of cases after 10 days (w)	340.29	678.64	30	1006
Number of cases after 20 days (w)	122218.17	311256.66	60	1006
Predicted mortality rate	15.60	21.47	5	1006

Notes: Online Appendix Table A.7 displays summary statistics for the experimental sample. These data were collected on March 16. Panel A shows shares of respondents with indicated characteristics. Panel B shows shares of respondents with indicated beliefs about the severity of the crisis and economic anxieties. Panel C shows variables measuring coronavirus perceptions and predictions of fictitious infectious disease spread.

Table A.8: US Opinion Polling Data on Reaction to Coronavirus From 13 - 18 Feb 2020

	(1)	(2)	(3)
	Strength of concern that		
	you/your family getting sick from coronavirus	coronavirus will have negative impact on US economy	widespread outbreak of coronavirus in US
Any Covid-19 case	0.111* (0.063)	0.124** (0.057)	0.123** (0.056)
Mean of DV	.00278	.00213	.000396
R ²	.179	.139	.186
States	50	50	50
Observations	1197	1192	1197
Individual Controls	X	X	X
Interview Date FE	X	X	X

Notes: Online Appendix Table A.8 presents results from studying the impact of the coronavirus on perceptions and awareness during the early period of the spread. Data come from an opinion poll conducted for the Kaiser Family Foundation Poll (Roper Center ID 31117209) from February 13 - February 18. During this time there were only 13 reported coronavirus cases in all of the US which were concentrated in 5 states. Outcomes are measured on a 4-point scale (Not at all concerned; Not too concerned; Somewhat concerned; Very concerned) and standardized to have mean zero and standard deviation one. Individual controls include age, gender, education, income and political party affiliation (5 point scale). Observation counts vary due to "don't knows" or non-response. Standard errors clustered at the state level are presented in parentheses with stars indicating * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.9: Coronavirus perceptions and economic anxieties over time

	March 5			March 16			Comparison of means	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Mean	SD	Obs.	Mean	SD	Obs.	Δ	p(early = late)
Panel A: Economic Anxieties								
% agree: world severely affected by coronavirus	80.12	39.93	1006	67.61	46.82	914	-12.50	0.000
% agree: US severely affected by coronavirus	77.83	41.56	1006	55.14	49.76	914	-22.69	0.000
% worried about US economy	87.57	33.00	1006	68.05	46.65	914	-19.52	0.000
% worried about personal econ. situation	73.76	44.02	1006	47.16	49.95	914	-26.60	0.000

Notes: Online Appendix Table A.9 displays summary statistics for economic anxieties (Panel A) and coronavirus perceptions (Panel B). Columns (1) - (3) display descriptives for Experiment 1 conducted on March 5, while Columns (4) to (6) display the descriptives for Experiment 2 conducted on March 16.

Table A.10: Impact of coronavirus arrival in US states on economic anxieties

	Predicted impact on (standardized)		Worry about (standardized)	
	(1) World	(2) US	(3) US Economy	(4) Pers. Economic Sit.
Any case	0.2318** (0.0985)	0.2579*** (0.0946)	0.2159* (0.1079)	0.0636 (0.1258)
R ²	0.07	0.07	0.08	0.09
Number of Observations	1920	1920	1920	1920
Including controls	X	X	X	X
State FE	X	X	X	X
Survey FE	X	X	X	X

Notes: Online Appendix Table A.10 displays the effect of having at least one confirmed coronavirus case on economic anxieties using a difference-in-differences estimation with survey data collected on March 5 and 16. All regressions include state and day fixed effects and control for gender, age bin dummies, log income, log income squared, dummies for having a high school degree and having some college education, dummies for being unemployed, currently working, a student and dummies for self-identifying as Democrat or Republican. The dependent variables in columns (1) to (2) are agreement on a five-point Likert-scale (from “strongly disagree” to “strongly agree”) with the statements “The world will be severely affected by the coronavirus.” (column (1)), and “The US will be severely affected by the coronavirus.” (column (2)). The dependent variables in columns (3) and (4) are answers on a four-point Likert-scale (from “not at all worried” to “very worried”) to the questions “Are you worried about the effects of the coronavirus on the US economy?” (column (3)) and “Are you worried about the effects of the coronavirus on your household’s economic situation?” (column (4)). All outcomes are standardized to have mean 0 and standard deviation 1 within each survey wave. Standard errors clustered at the state level are presented in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.11: The association of misperceptions and economic anxieties

	Predicted impact on (standardized)		Worry about (standardized)	
	(1) World	(2) US	(3) US Economy	(4) Pers. Economic Sit.
Panel A: No control variables				
Overestimate mortality	0.3655*** (0.0647)	0.4516*** (0.0637)	0.1892*** (0.0655)	0.4783*** (0.0635)
Overestimate contagiousness	0.5263*** (0.0899)	0.5722*** (0.0843)	0.4504*** (0.0879)	0.4096*** (0.0825)
R ²	0.08	0.11	0.04	0.09
Panel B: Including control variables				
Overestimate mortality	0.3830*** (0.0654)	0.4390*** (0.0642)	0.1850*** (0.0668)	0.4016*** (0.0645)
Overestimate contagiousness	0.5077*** (0.0885)	0.5392*** (0.0832)	0.4252*** (0.0849)	0.3780*** (0.0810)
R ²	0.13	0.16	0.10	0.16
Number of Observations	914	914	914	914

Notes: Online Appendix Table A.11 displays the raw effect of overestimating mortality and contagiousness of coronavirus (relative to official estimates) on the perceived severity of the effects of the coronavirus and economic worries. The data were collected on March 5. Panel A shows the results without control variables. Panel B shows the results controlling for gender, age bin dummies, log income, log income squared, dummies for having a high school degree and having some college education, dummies for being unemployed, currently working, a student and dummies for self-identifying as Democrat or Republican. The table shows coefficients estimated using linear regressions that include indicators for respondents whose beliefs about coronavirus mortality were higher relative to official estimates and for respondents whose beliefs about coronavirus contagiousness were higher relative to scientific estimates. The dependent variables in columns (1) to (2) are agreement on a five-point Likert-scale (from “strongly disagree” to “strongly agree”) with the statements “The world will be severely affected by the coronavirus.” (column (1)), and “The US will be severely affected by the coronavirus.” (column (2)). The dependent variables in columns (3) and (4) are answers on a four-point Likert-scale (from “not at all worried” to “very worried”) to the questions “Are you worried about the effects of the coronavirus on the US economy?” (column (3)) and “Are you worried about the effects of the coronavirus on your household’s economic situation?” (column (4)). All outcomes are standardized to have mean 0 and standard deviation 1. Heteroskedasticity robust standard errors are presented in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.12: The association of misperceptions and economic anxieties - continuous measures

	Predicted impact on (standardized)		Worry about (standardized)	
	(1) World	(2) US	(3) US Economy	(4) Pers. Economic Sit.
Panel A: No control variables				
Perceived mortality	0.8220*** (0.1887)	1.1333*** (0.1730)	0.6213*** (0.2013)	1.3450*** (0.1841)
Perceived contagiousness	0.2714** (0.1243)	0.6121*** (0.1071)	0.4826*** (0.1200)	0.5845*** (0.1149)
R ²	0.03	0.09	0.04	0.11
Panel B: Including control variables				
Perceived mortality	0.8841*** (0.1932)	1.1090*** (0.1804)	0.6747*** (0.2098)	1.1382*** (0.1919)
Perceived contagiousness	0.2728** (0.1239)	0.6084*** (0.1070)	0.4810*** (0.1226)	0.5596*** (0.1187)
R ²	0.08	0.14	0.10	0.17
Number of Observations	914	914	914	914

Notes: Online Appendix Table A.12 displays the correlation between mortality and contagiousness of coronavirus on the perceived severity of the effects of the coronavirus and economic worries. The data were collected on March 5. Panel A shows the results without control variables. Panel B shows the results controlling for gender, age bin dummies, log income, log income squared, dummies for having a high school degree and having some college education, dummies for being unemployed, currently working, a student and dummies for self-identifying as Democrat or Republican. The table shows coefficients for beliefs about coronavirus mortality and about coronavirus contagiousness estimated using linear regressions. Beliefs about mortality are rescaled to range from 0 to 1. Beliefs about contagiousness are rescaled to display the association with believing that 100 extra people get infected. Beliefs about mortality and contagiousness are winsorized at the 95th percentile to account for outliers. The dependent variables in columns (1) to (2) are agreement on a five-point Likert-scale (from “strongly disagree” to “strongly agree”) with the statements “The world will be severely affected by the coronavirus.” (column (1)), and “The US will be severely affected by the coronavirus.” (column (2)). The dependent variables in columns (3) and (4) are answers on a four-point Likert-scale (from “not at all worried” to “very worried”) to the questions “Are you worried about the effects of the coronavirus on the US economy?” (column (3)) and “Are you worried about the effects of the coronavirus on your household’s economic situation?” (column (4)). All outcomes are standardized to have mean 0 and standard deviation 1. Heteroskedasticity robust standard errors are presented in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.13: Experimental integrity: Balance table

	Mortality information experiment			Contagion information experiment		
	(1) Mean low rel. mortality	(2) Mean high rel. mortality	(3) p(low rel. mort. = high rel. mort)	(4) Mean no contagion info	(5) Mean contagion info	(6) p(no info = info)
% Male	50.55	47.49	0.36	50.11	47.93	0.51
% Age < 35	23.74	24.62	0.76	23.96	24.40	0.88
% Highschool education	18.90	16.34	0.31	16.70	18.52	0.47
% College education	78.90	82.14	0.22	81.98	79.08	0.27
% Currently working	58.46	51.63	0.04	55.38	54.68	0.83
% Democrat	38.90	41.18	0.48	37.58	42.48	0.13
% Republican	33.41	32.90	0.87	34.95	31.37	0.25
% High trust in science	1.98	0.22	0.01	0.88	1.31	0.53
p-value of joint significance			0.00			0.74

Notes: Online Appendix Table A.13 displays balance tests for the experimental sample. The data were collected on March 5. Columns (1) to (3) show means for both experimental groups in the mortality information experiment and the p-value for a test of equality of means across samples. Columns (4) to (6) show means for both experimental groups in the contagiousness experiment and the p-value for a test of equality of means across samples. p-values are obtained using heteroskedasticity robust standard errors. p-values for the test of joint significance are based on the F-statistic obtained by regressing all observables on the treatment indicators.

Table A.14: The impact of coronavirus-related information on perceived severity of the crisis

	Predicted impact on (standardized)	
	(1) World	(2) US
Panel A: No control variables		
High relative mortality	0.2833*** (0.0655)	0.2246*** (0.0658)
R ²	0.02	0.01
Number of Observations	914	914
Panel B: Including control variables		
High relative mortality	0.2705*** (0.0659)	0.2245*** (0.0651)
R ²	0.06	0.07
Number of Observations	914	914

Notes: Online Appendix Table A.14 displays the impact of information about the coronavirus on the perceived severity of the impacts of the coronavirus. The data were collected on March 5. Panel A shows the results without control variables. Panel B shows the results controlling for gender, age bin dummies, log income, log income squared, dummies for having a high school degree and having some college education, dummies for being unemployed, currently working, a student and dummies for self-identifying as Democrat or Republican. The table shows coefficients estimated using linear regressions that compare respondents who were truthfully informed that the death rate of the coronavirus is either “20 times higher than for the flu” (high mortality treatment), or “5 times lower than for SARS” (low mortality treatment). The dependent variables in columns (1) to (2) are agreement on a five-point Likert-scale (from “strongly disagree” to “strongly agree”) with the statements “The world will be severely affected by the coronavirus.” (column (1)), and “The US will be severely affected by the coronavirus.” (column (2)). All outcomes are standardized to have mean 0 and standard deviation 1. Heteroskedasticity robust standard errors are presented in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.15: The impact of coronavirus-related information on economic worries

	Worry about (standardized)	
	(1) US Economy	(2) Pers. Economic Sit.
Panel A: No control variables		
High relative mortality	0.1557** (0.0660)	0.1560** (0.0660)
R ²	0.01	0.01
Number of Observations	914	914
Contagion information	-0.0116 (0.0434)	-0.0866** (0.0415)
R ²	0.57	0.61
Number of Observations	914	914
Panel B: Including control variables		
High relative mortality	0.1556** (0.0660)	0.1834*** (0.0645)
R ²	0.07	0.10
Number of Observations	914	914
Contagion information	-0.0064 (0.0435)	-0.0826** (0.0416)
R ²	0.58	0.62
Number of Observations	914	914

Notes: Online Appendix Table A.15 displays the impact of information about the coronavirus on economic anxiety. The data were collected on March 5. The results show that information about coronavirus causally affects economic anxiety. Panel A shows the results without control variables. Panel B shows the results controlling for gender, age bin dummies, log income, log income squared, dummies for having a high school degree and having some college education, dummies for being unemployed, currently working, a student and dummies for self-identifying as Democrat or Republican. *High relative mortality* shows coefficients estimated using linear regressions that compare respondents who were truthfully informed that the death rate of the coronavirus is either “20 times higher than for the flu” (high mortality treatment), or “5 times lower than for SARS” (low mortality treatment). *Contagion information* shows regression coefficients that compare respondents who were truthfully informed about the estimated contagiousness of coronavirus ($R_0 \approx 2$) to respondents who were given no information. Estimates for *Contagion information* are obtained with an ANCOVA specification using baseline outcomes obtained in the same survey prior to the information treatment. The dependent variables in columns (1) and (2) are answers on a four-point Likert-scale (from “not at all worried” to “very worried”) to the questions “Are you worried about the effects of the coronavirus on the US economy?” (column (1)) and “Are you worried about the effects of the coronavirus on your household’s economic situation?” (column (2)). All outcomes are standardized to have mean 0 and standard deviation 1. Heteroskedasticity robust standard errors are presented in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.16: Controlling for treatment assignment in the relative mortality experiment

	Worry about (standardized)	
	(1) US Economy	(2) Pers. Economic Sit.
Panel A: No control variables		
Contagion information	-0.0107 (0.0435)	-0.0876** (0.0414)
R ²	0.57	0.61
Number of Observations	914	914
Panel B: Including control variables		
Contagion information	-0.0052 (0.0436)	-0.0836** (0.0415)
R ²	0.58	0.62
Number of Observations	914	914

Notes: Online Appendix Table A.16 displays the impact of information on the contagiousness of coronavirus on economic anxiety. The data were collected on March 5. All specifications include a dummy for whether the respondent saw the high relative mortality treatment during the first experimental variation. Panel A shows the results without control variables. Panel B shows the results controlling for gender, age bin dummies, log income, log income squared, dummies for having a high school degree and having some college education, dummies for being unemployed, currently working, a student and dummies for self-identifying as Democrat or Republican. *Contagion information* shows regression coefficients that compare respondents who were truthfully informed about the estimated contagiousness of coronavirus (R0≈2) to respondents who were given no information. Estimates are obtained with an ANCOVA specification using baseline outcomes obtained in the same survey prior to the information treatment. The dependent variables in columns (1) and (2) are answers on a four-point Likert-scale (from “not at all worried” to “very worried”) to the questions “Are you worried about the effects of the coronavirus on the US economy?” (column (1)) and “Are you worried about the effects of the coronavirus on your household’s economic situation?” (column (2)). All outcomes are standardized to have mean 0 and standard deviation 1. Heteroskedasticity robust standard errors are presented in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.17: Interaction effects between contagion and mortality information

	Worry about (standardized)	
	(1) US Economy	(2) Pers. Economic Sit.
Panel A: No control variables		
Contagion information	0.0270 (0.0576)	-0.1243** (0.0493)
Contagion information $\times$ Low relative mortality	-0.0759 (0.0675)	0.0743 (0.0571)
Baseline value	0.7534*** (0.0248)	0.7803*** (0.0228)
R ²	0.57	0.61
Number of Observations	914	914
Panel B: Including control variables		
Contagion information	0.0427 (0.0576)	-0.1188** (0.0494)
Contagion information $\times$ Low relative mortality	-0.0965 (0.0667)	0.0709 (0.0567)
Baseline value	0.7467*** (0.0259)	0.7572*** (0.0250)
R ²	0.58	0.62
Number of Observations	914	914

Notes: Online Appendix Table A.17 shows the interaction effect of our treatments on economic anxiety. The data were collected on March 5. Panel A shows the results without control variables. Panel B shows the results controlling for gender, age bin dummies, log income, log income squared, dummies for having a high school degree and having some college education, dummies for being unemployed, currently working, a student and dummies for self-identifying as Democrat or Republican. The dependent variables in columns (1) and (2) are answers on a four-point Likert-scale (from “not at all worried” to “very worried”) to the questions “Are you worried about the effects of the coronavirus on the US economy?” (column (1)) and “Are you worried about the effects of the coronavirus on your household’s economic situation?” (column (2)). The outcome variables are elicited after both treatments. Estimates are obtained with an ANCOVA specification using baseline outcomes obtained in the same survey after the mortality treatment and prior to the contagiousness information treatment. We do not include a baseline indicator for the mortality treatment as it would be collinear with the baseline measures of economic worries which are themselves sufficient statistics for the assignment probabilities regarding the mortality treatments. For interpretation, we use the interaction between the low mortality treatment (as opposed to the high mortality treatment), as it is complementary to the contagiousness information given that both treatments render the severity of the coronavirus less intense. All outcomes are standardized to have mean 0 and standard deviation 1. Heteroskedasticity robust standard errors are presented in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.18: Predictors of mental models of disease spread

	Log predicted cases after twenty days		Exponential mental model	
	(1) OLS	(2) Post LASSO	(3) OLS	(4) Post LASSO
Male	0.236 (0.287)	0.213 (0.285)	0.0163 (0.0240)	
Aged 25 to 34	-0.0312 (0.490)	-0.244 (0.381)	0.000318 (0.0415)	-0.0361 (0.0300)
Aged 35 to 44	0.153 (0.531)	-0.115 (0.407)	0.0539 (0.0455)	
Aged 45 to 54	0.296 (0.495)		-0.0193 (0.0398)	-0.0603** (0.0272)
Aged 55 to 64	0.536 (0.534)	0.354 (0.421)	0.0400 (0.0431)	
Aged 65 above	1.289** (0.560)	1.097** (0.450)	0.138*** (0.0472)	0.106*** (0.0353)
Log income	2.792 (3.450)	0.370** (0.182)	0.387 (0.271)	0.0342** (0.0139)
Log income squared	-0.113 (0.162)		-0.0166 (0.0127)	
Highschool education	1.265** (0.499)		0.0793** (0.0370)	
College education	2.257*** (0.477)	1.184*** (0.314)	0.144*** (0.0354)	0.0781*** (0.0255)
Currently working	-0.687* (0.379)	-0.667* (0.373)	-0.0428 (0.0322)	-0.0366 (0.0278)
Student	-0.804 (0.602)	-0.780 (0.601)	-0.0317 (0.0472)	
Democrat	-0.658** (0.330)	-0.642* (0.329)	-0.0420 (0.0278)	-0.0346 (0.0230)
Republican	-0.359 (0.376)	-0.336 (0.376)	-0.0165 (0.0302)	
Constant	-12.79 (18.34)	1.335 (1.764)	-2.215 (1.436)	-0.250* (0.140)
R ²	0.0517	0.0485	0.0559	0.0510
Number of observations	1006	1006	1006	1006

Notes: Online Appendix Table A.18 displays the correlations of covariates with different classifications of mental models. These data were collected on March 16. Columns (1) and (2) show the correlation with log predicted of cases after 20 days from a fictitious disease. Columns (3) and (4) show the correlation with an indicator variable being classified as having an exponential model by k-means clustering. Columns (1) and (3) show the result for a simple OLS regression. Columns (2) and (4) show results using OLS with only the variables selected by the LASSO algorithm using α and λ parameters chosen by 10-fold cross-validation. Heteroskedasticity robust standard errors are presented in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.19: The association of mental models of infectious disease spread and economic anxieties

	Predicted impact on (standardized)				Worry about (standardized)			
	(1) World	(2) World	(3) US	(4) US	(5) US Economy	(6) US Economy	(7) Pers. Economic Sit.	(8) Pers. Economic Sit.
Panel A								
Log(estimate day 5)- z-score	0.0971*** (0.0284)	0.0832*** (0.0282)	0.0942*** (0.0299)	0.0893*** (0.0305)	0.0747** (0.0306)	0.0750** (0.0306)	0.0044 (0.0318)	0.0268 (0.0325)
R ²	0.01	0.07	0.01	0.06	0.01	0.06	0.00	0.09
Number of Observations	1006	1006	1006	1006	1006	1006	1006	1006
Panel B								
Log(estimate day 10)- z-score	0.1044*** (0.0291)	0.0857*** (0.0292)	0.1137*** (0.0295)	0.1069*** (0.0303)	0.0714** (0.0313)	0.0703** (0.0314)	-0.0229 (0.0327)	0.0086 (0.0321)
R ²	0.01	0.07	0.01	0.06	0.01	0.06	0.00	0.09
Number of Observations	1006	1006	1006	1006	1006	1006	1006	1006
Panel C								
Log(estimate day 20)- z-score	0.0828*** (0.0313)	0.0608* (0.0315)	0.0933*** (0.0314)	0.0841*** (0.0323)	0.0502 (0.0324)	0.0489 (0.0326)	-0.0441 (0.0332)	-0.0079 (0.0327)
R ²	0.01	0.06	0.01	0.06	0.00	0.06	0.00	0.09
Number of Observations	1006	1006	1006	1006	1006	1006	1006	1006
Controls	X			X	X			X

Notes: Online Appendix Table A.19 displays the regression coefficients of perceived severity of the effects of the coronavirus with participants' standardized log estimate of the spread of a fictitious disease. The data were collected on March 16. The table shows coefficients estimated using linear regressions that regress perceived crisis severity and economic anxieties on the z-scored log of estimated infections from a fictitious disease. The dependent variables in columns (1) to (4) are agreement on a five-point Likert-scale (from "strongly disagree" to "strongly agree") with the statements "The world will be severely affected by the coronavirus." (columns (1) and (2)), and "The US will be severely affected by the coronavirus." (columns (3) and (4)). The dependent variables in columns (5) to (8) are answers on a four-point Likert-scale (from "not at all worried" to "very worried") to the questions "Are you worried about the effects of the coronavirus on the US economy?" (columns (5) and (6)) and "Are you worried about the effects of the coronavirus on your household's economic situation?" (columns (7) and (8)). The right-hand-side variables are the standardized log of participants' estimates for the number of people infected with the fictitious disease on day 5, day 10 and day 20, respectively. All outcomes are standardized to have mean 0 and standard deviation 1. Even columns control for gender, age bin dummies, log income, log income squared, dummies for having a high school degree and having some college education, dummies for being unemployed, being currently working, being a student and dummies for self-identifying as Democrat and Republican. Heteroskedasticity robust standard errors are presented in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$